{EPITECH}

<SINCE 1999/>

< Robotique TEK5 · Online class />

Robotics & AI: hype vs reality

{ <TECH TOGETHER TOMORROW> }

< Today />

PART 1 · ~55 MIN

Robotics culture (Reloaded)

- NVIDIA & Hugging Face
- Models & world models
- Fake autonomy
- Europe & defense
- Ethics: jobs, energy, war

PART 2 · ~30 MIN

Your projects

- Phase 2: 13 groups, 5 themes
- Stats
- Follow-ups: 10-min slots

{EPITECH}

<SINCE 1999/>

< Part 1 />

Part 1: hype vs reality

{ <TECH TOGETHER TOMORROW> }

{EPITECH}

<SINCE 1999/>

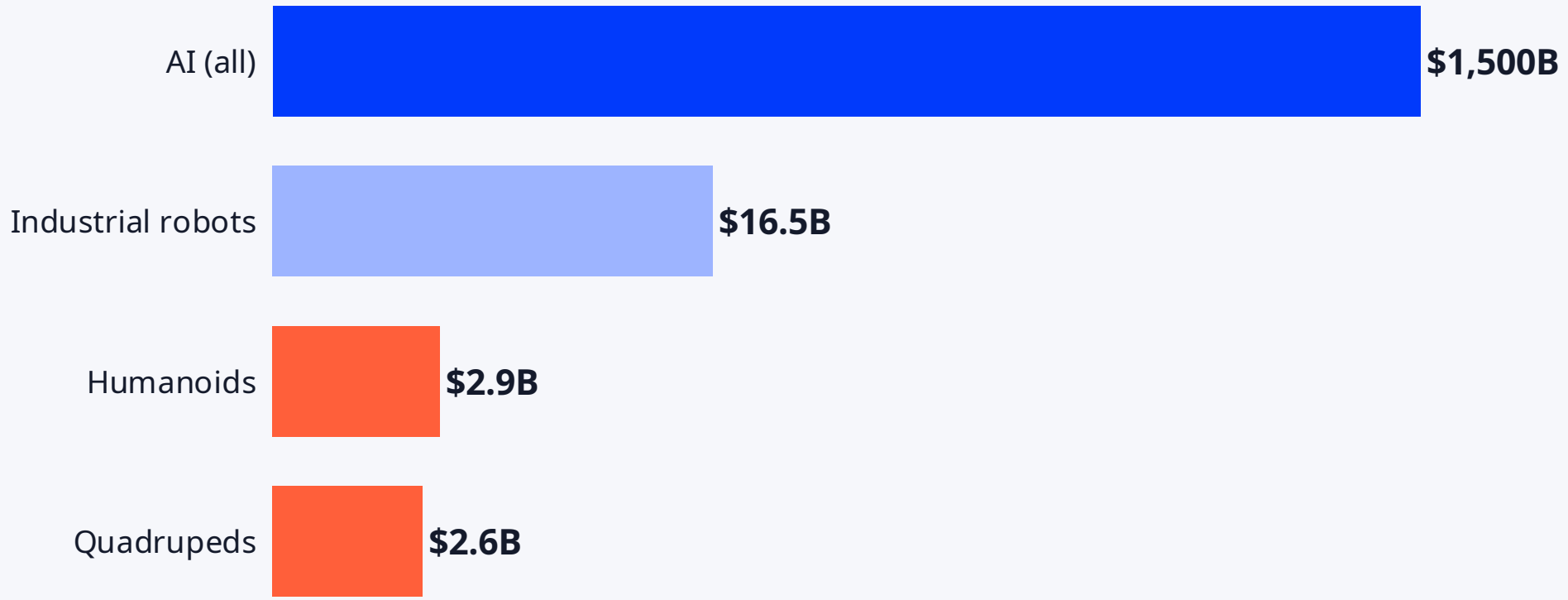
< Part 1 · 1/5 />

Robotics industry
NVIDIA & Hugging Face

{ <TECH TOGETHER TOMORROW> }

< The money today: AI vs robots />

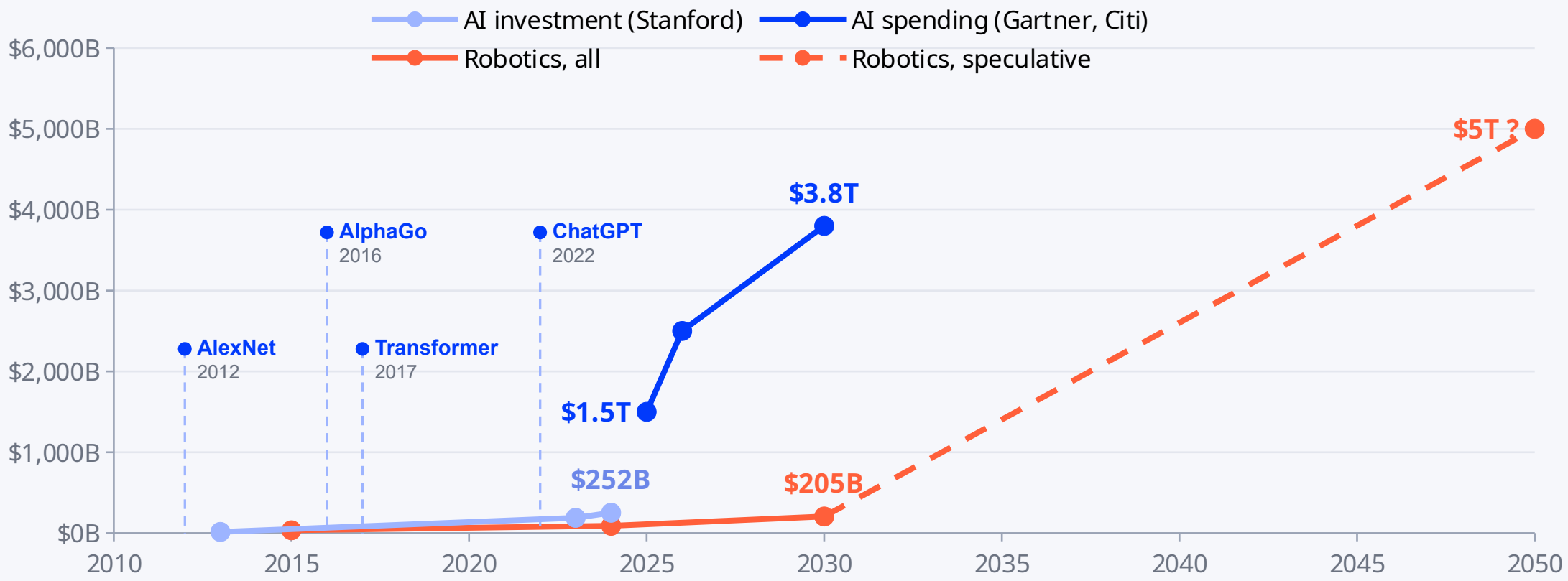
AI ≈ 70× all these robots combined



Sources: Gartner (AI spending 2025) · IFR World Robotics (industrial robot installations, hardware only) · MarketsandMarkets (humanoids) · market research est. (quadrupeds) · log scale

< Tomorrow: the forecasts />

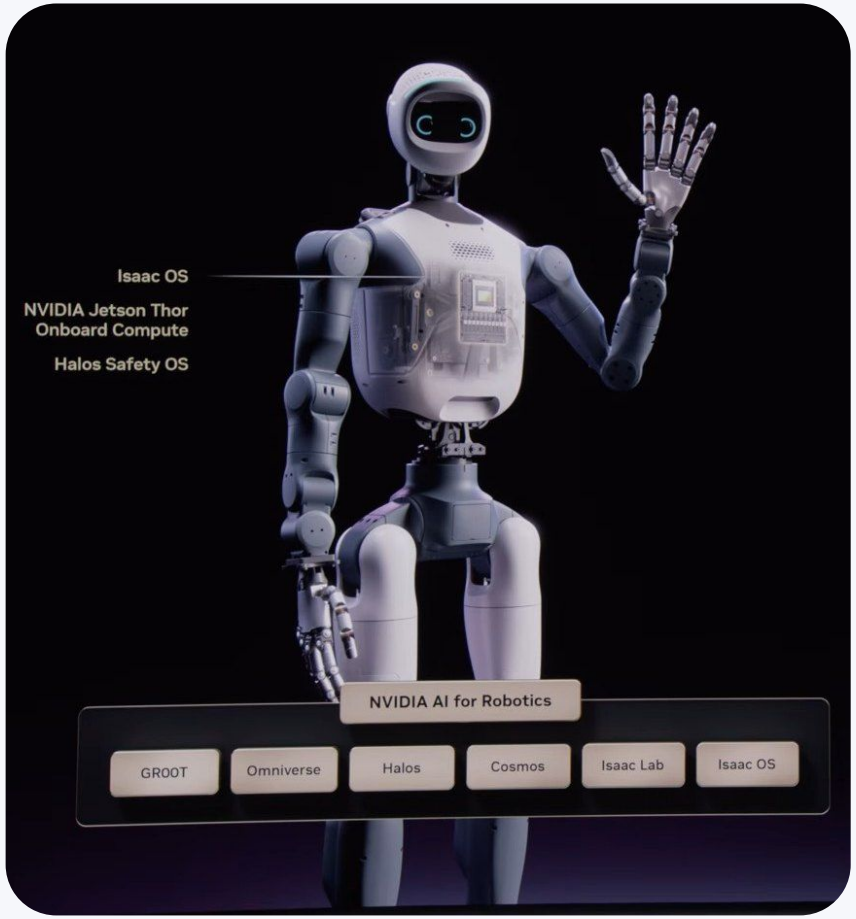
Robotics: $90B today, $5T in 2050?



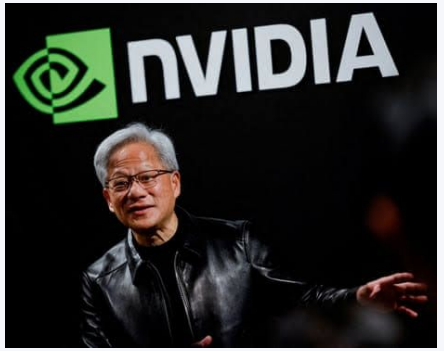
Sources: AI: Stanford AI Index (corporate investment, 2013-24), Gartner (spending, 2025-26), Citi (2030) · Robotics: IFR (2015), GlobalData (2024-30), Morgan Stanley humanoids (2050)

Would you bet on the dashed line?

< Physical AI: NVIDIA's recipe />



Sources: NVIDIA GTC Berlin keynote (Oct. 2026), from 1:05:19



Three computers

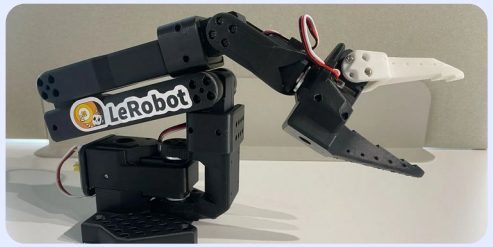
1 Train
DGX, in the cloud

2 Simulate
Omniverse · Isaac Sim · Isaac Lab · Newton

3 Run
Jetson Thor on the robot + safety OS

Reference robot: NVIDIA Agile One (with Agile Robots), Q1 2027

< Hugging Face: a robot lab for the price of a phone />



$100

SO-101

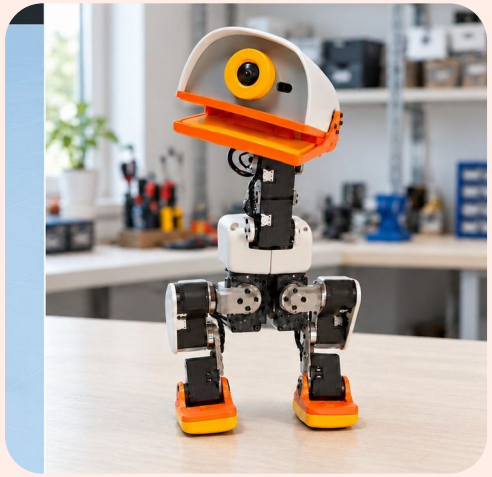
the arm on your desk



$300

Reachy Mini

desktop HRI robot



$399

Microduck

RL-trained biped duck



$70k too expensive for you

Reachy 2

open-source humanoid

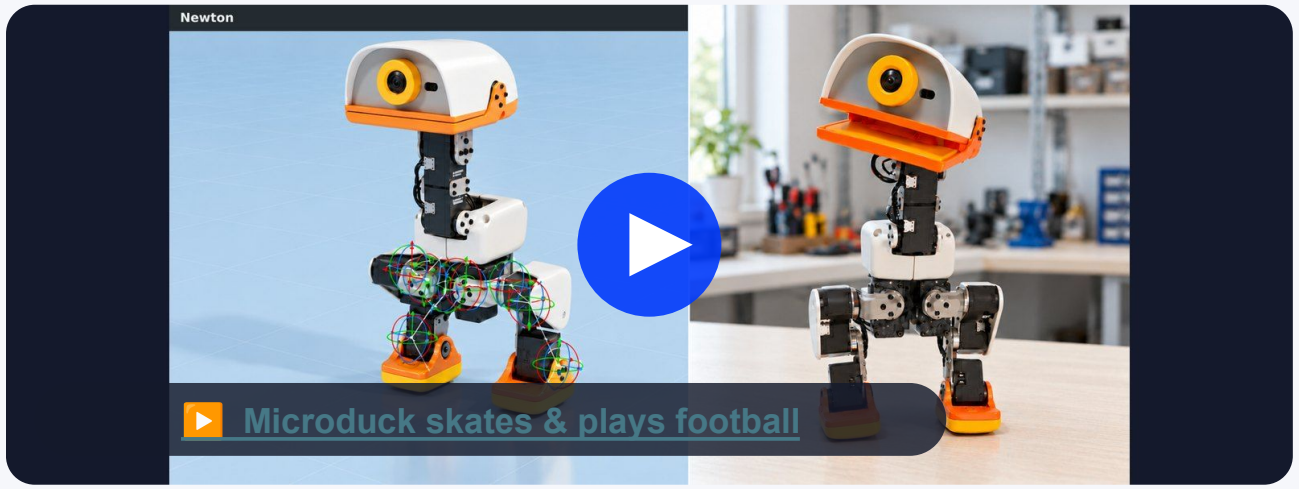
Sources: Hugging Face · Pollen Robotics (Reachy 2, $70k) · TechCrunch (May 2025) · Axios (Aug. 2026)

< The Microduck challenge />

ANTOINE

#?

on the leaderboard
RL in sim → real duck

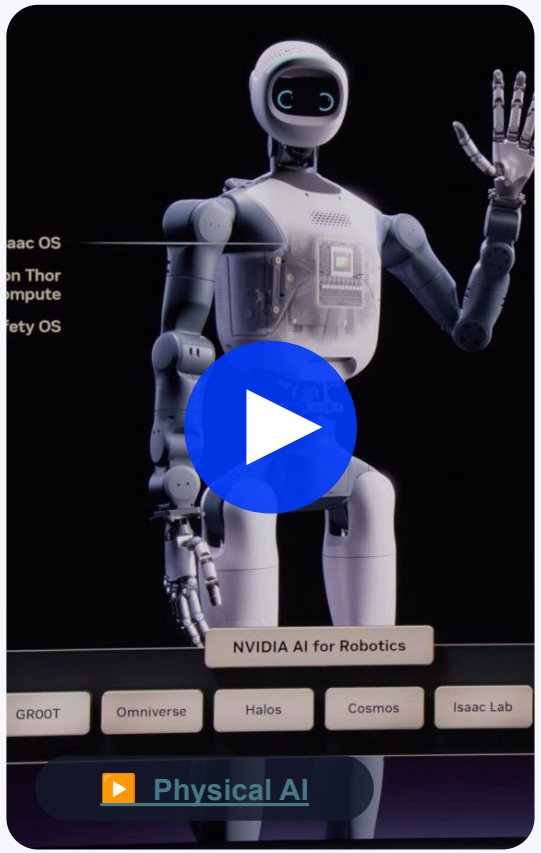


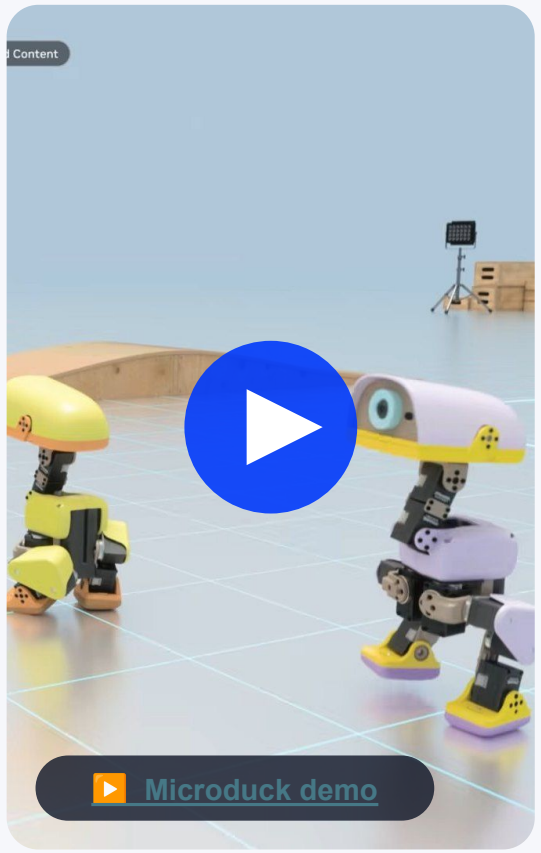
SITE · Leaderboard

add the challenge link

< GTC Berlin, Oct. 2026: Jensen in Europe />







Simulation first, real robot last. Why?

{EPITECH}

<SINCE 1999/>

< Part 1 · 2/5 />

AI Models & world models

{ <TECH TOGETHER TOMORROW> }

< Not everything needs an LLM />



“Is the red cube in the gripper?”

LLM API

writes a sentence

> Yes, the red cube appears to be held firmly between the two fingers of the gripper, although the angle…

3–329 s
latency

$7.50
per 1,000 questions

chat · reasoning

Clef-Flash

Cloudflare · open weights · 9B

`input: camera image`

```
grasped: yes  ·  p = 0.97
```

39 ms
latency

$0.05
per 1,000 questions

grasp check · scene check

Jev

TypeSafe AI · closed · text only

`input: “gripper=closed, object=red_cube”`

```
grasped: yes  ·  p = 0.97
```

70–500 ms
latency

$0.02
per 1,000 questions

routing · scoring

System 1 models: one pass → a typed answer (yes/no, choice, score), not text.

Sources: TypeSafe AI (Jev, Sept. 2026) · Better Stack (Jev vs Clef, Oct. 2026) · cost = 500 input tokens per question

< What does one robot decision cost? />

Your SO-101: 10 decisions/s for 1 hour = 36,000 calls × ~550 tokens

MODEL	LATENCY	FITS 10 Hz?	REAL MONEY / HOUR (log)
Frontier LLM API	3–329 s	✗	**$270**
Clef (Cloudflare, open)	209 ms	✗	**$4.30**
Clef-Flash (open, 9B)	39 ms	✓	**$1.60**
Jev (text only)	70–500 ms	✗	**$0.76**
SmolVLA on your GPU	local	✓	**≈ €0.09**

A humanoid like Figure's: fast control loop at 200 Hz = 720,000 calls per hour

Frontier LLM API	3–329 s	✗	**$5,400**
On-board Jetson Thor	local	✓	**≈ €0.03**

Sources: TypeSafe AI · Better Stack (Clef) · Figure Helix (200 Hz) · Jetson AGX Thor ≤130 W · image ≈ w×h/750 tokens (Anthropic) · electricity €0.25/kWh, my estimates

< A neural network that IS Rocket League />

5B

parameters

4

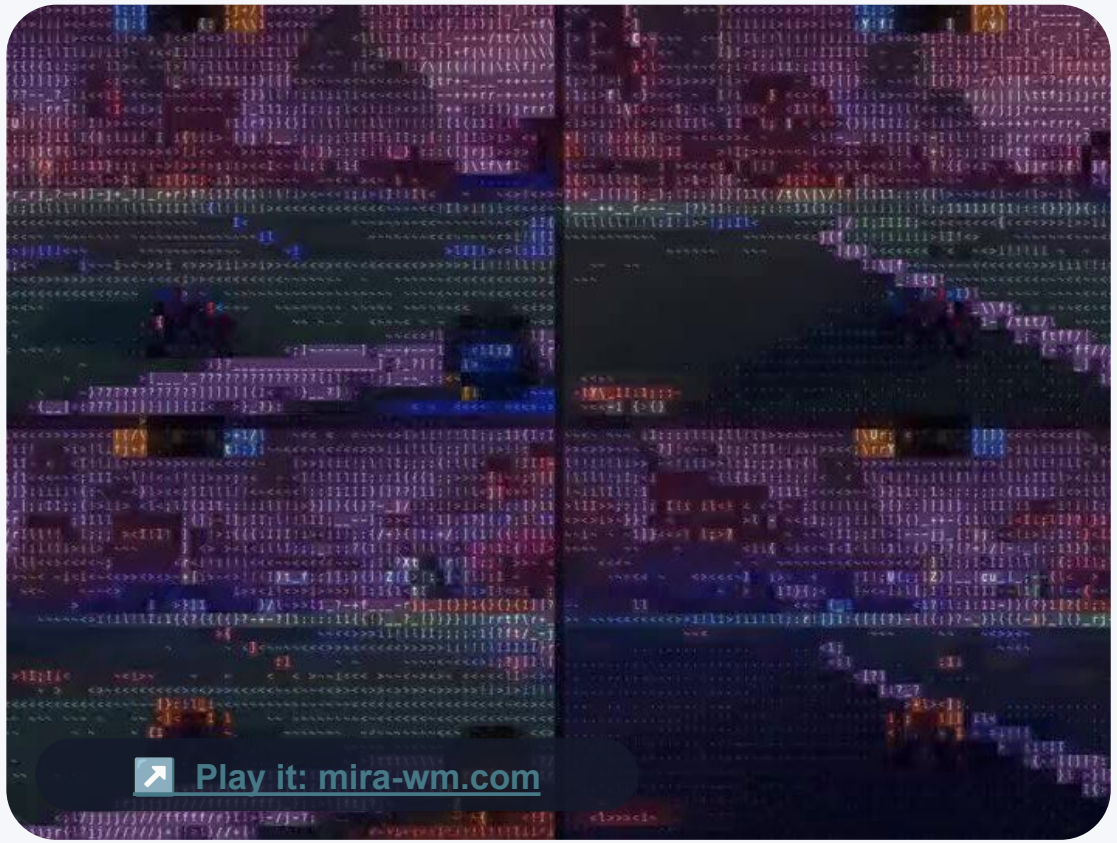
players live

20 fps

real time

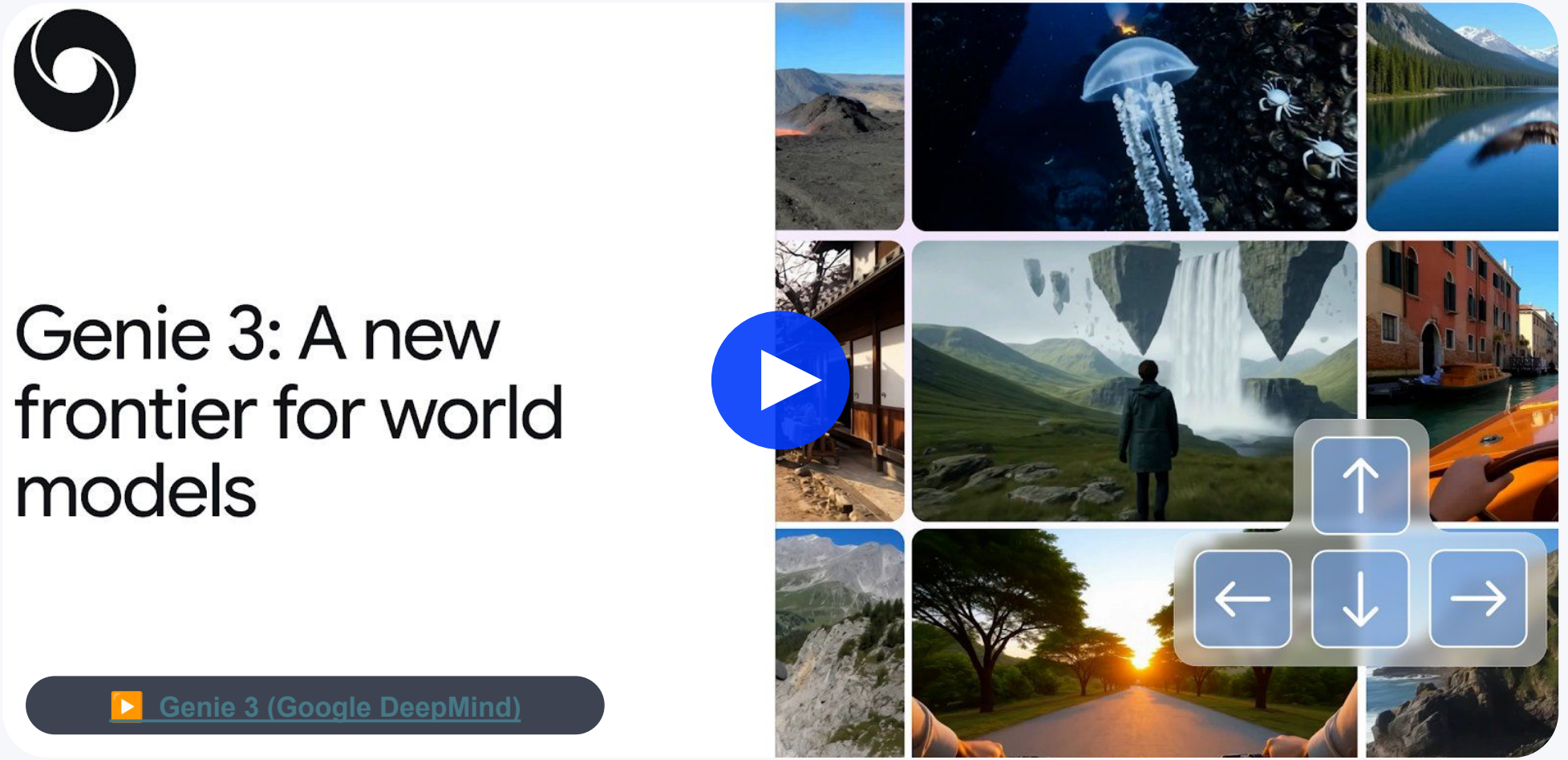
1

B200 GPU

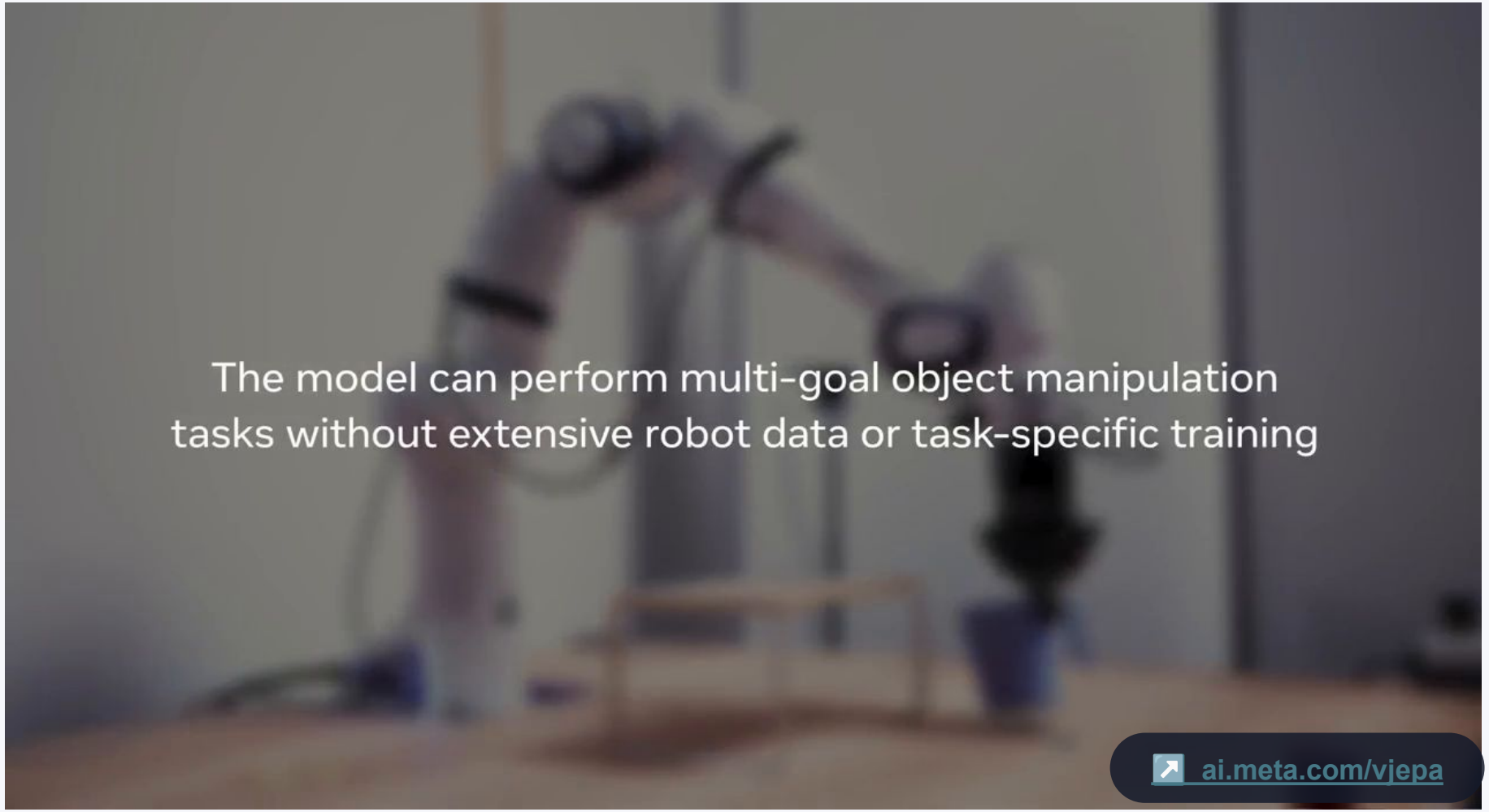


Sources: MIRA, arXiv 2607.05352 (July 2026)

< Genie 3: from video games to the real world />



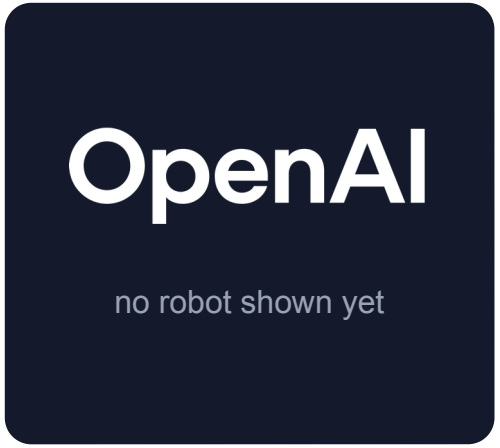
< JEPA: predict the meaning, not the pixels />



Trained on video, then 62 h of robot data → plans pick & place in labs it never saw

Source: Meta FAIR, V-JEPA 2 (June 2025) · ai.meta.com/vjepa

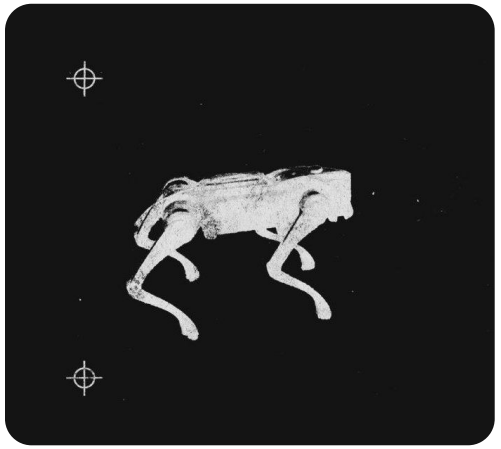
< LLM labs go physical />



OpenAI

In-house humanoid
“cool demo in 2027”

~27 robotics jobs



Anthropic

Claude codes a robot dog
~20× faster than humans

Project Fetch

Google DeepMind

Gemini Robotics
VLA + reasoning

closed, API only



Mistral

Robostral Navigate
one camera, no LiDAR

the European bet

Sources: Fortune / Let’s Data Science (OpenAI, Oct. 2026) · Anthropic Project Fetch phase 2 (June 2026)

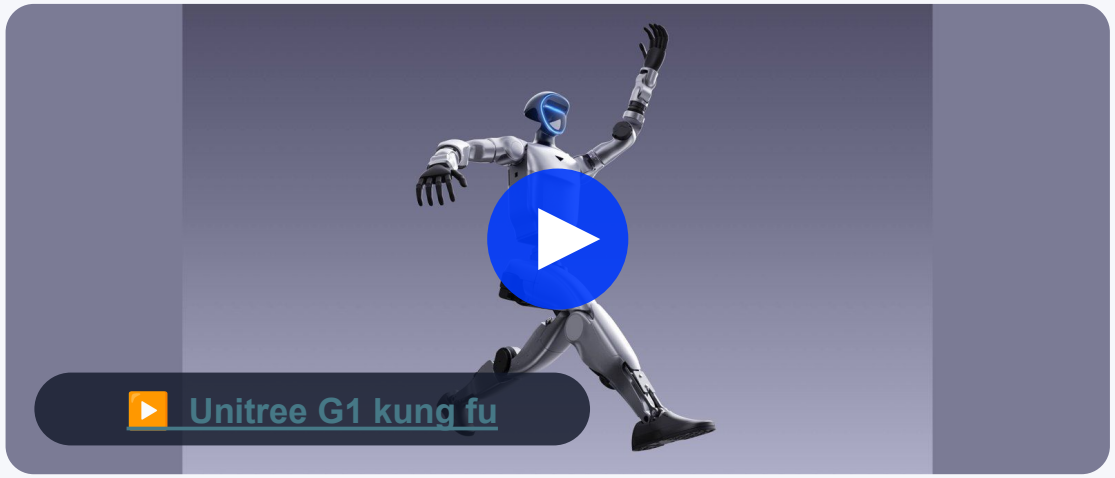
{EPITECH}

<SINCE 1999/>

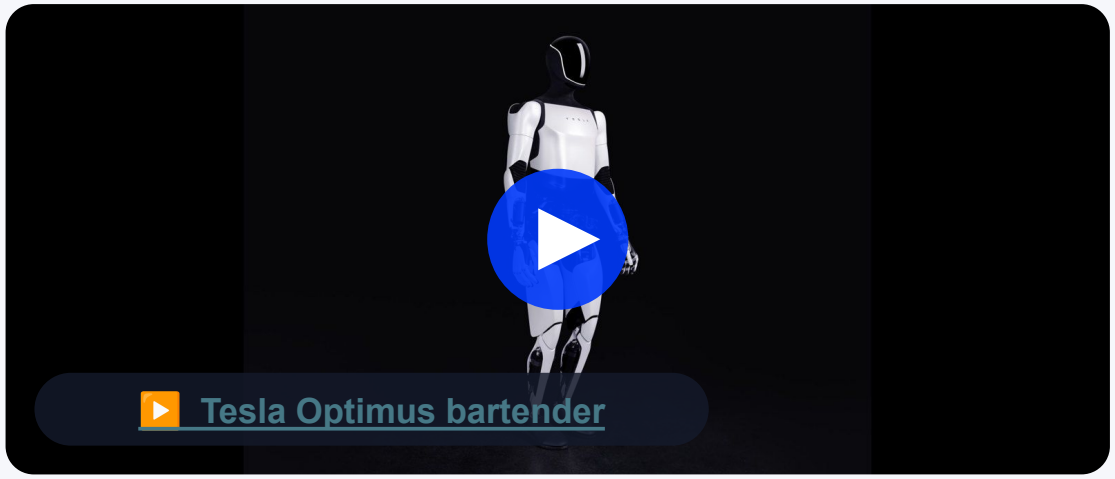
< Part 1 · 3/5 />

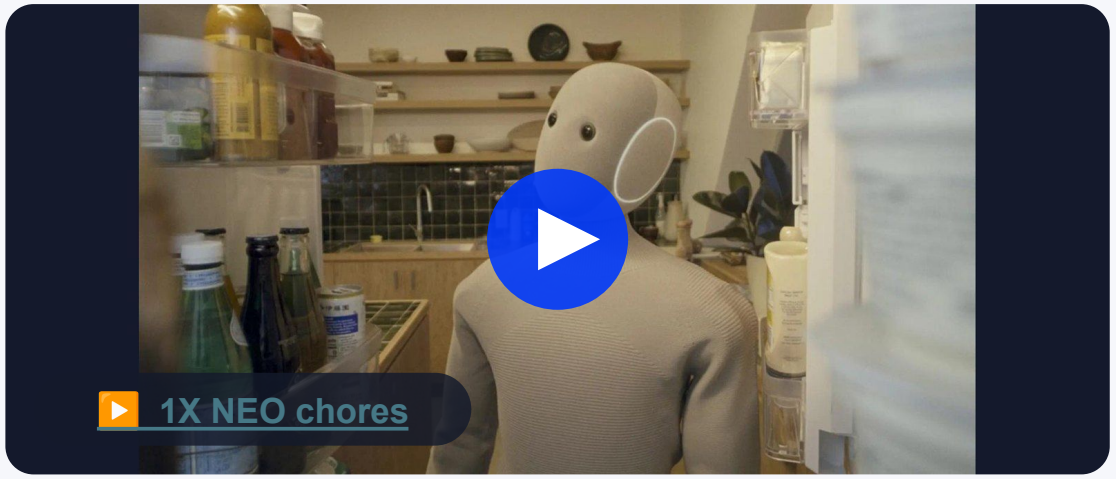
Fake autonomy

< Quiz: really autonomous? />









TEAMS POLL › How many are autonomous? 0 / 1 / 2 / 3 / 4

< Answer: 0 out of 4 decide by themselves />



Unitree G1 kung fu

Human motion captured, then replayed

NOT AUTONOMOUS



Unitree dancing

Fixed choreography, replayed

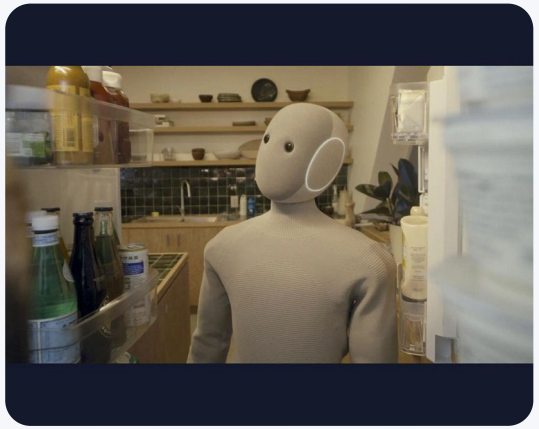
NOT AUTONOMOUS



Tesla Optimus bartender

Teleoperated (Bloomberg)

NOT AUTONOMOUS



1X NEO chores

Remote operator in VR (WSJ)

NOT AUTONOMOUS

Learned for real: balance & walking (RL in simulation). Not learned: deciding what to do.

< How it is really done: a human in a suit />

The operator kicks.
The robot kicks.
The operator…

Unitree G1 + motion-capture suit (Dec. 2025)
The robot copies the human, frame by frame

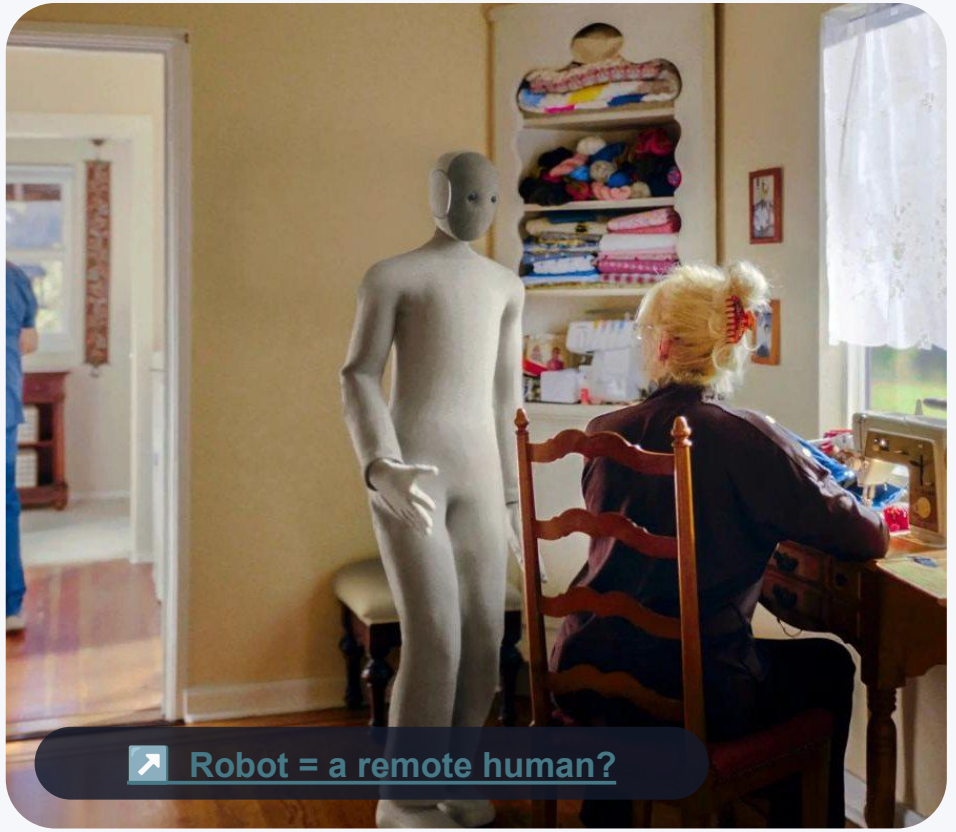


Sources: Interesting Engineering (Dec. 2025)

< A stranger in your living room />

A human operator.
A VR headset.
Your home.

Home humanoids learn from teleop data:
camera + mic + a remote human



Sources: InsideEVs / WSJ review of 1X NEO · BGR

Would you accept it for a $499/month robot?

{EPITECH}

<SINCE 1999/>

< Part 1 · 4/5 />

Europe & defense

{ <TECH TOGETHER TOMORROW> }

< Le Chonk: sovereign AI, made in Europe />





Mistral Large 4,
“very officially: le Chonk”

Trained in Europe

3,800 Grace Blackwell GPUs in Mistral’s own data centres

1T params, open weights

MoE, 49B active · 1M context · 160+ languages, all EU ones

Best non-Chinese open model

Artificial Analysis index 38 · closed frontier: 53–58

Runs under EU law

European deployment operated end-to-end by Mistral

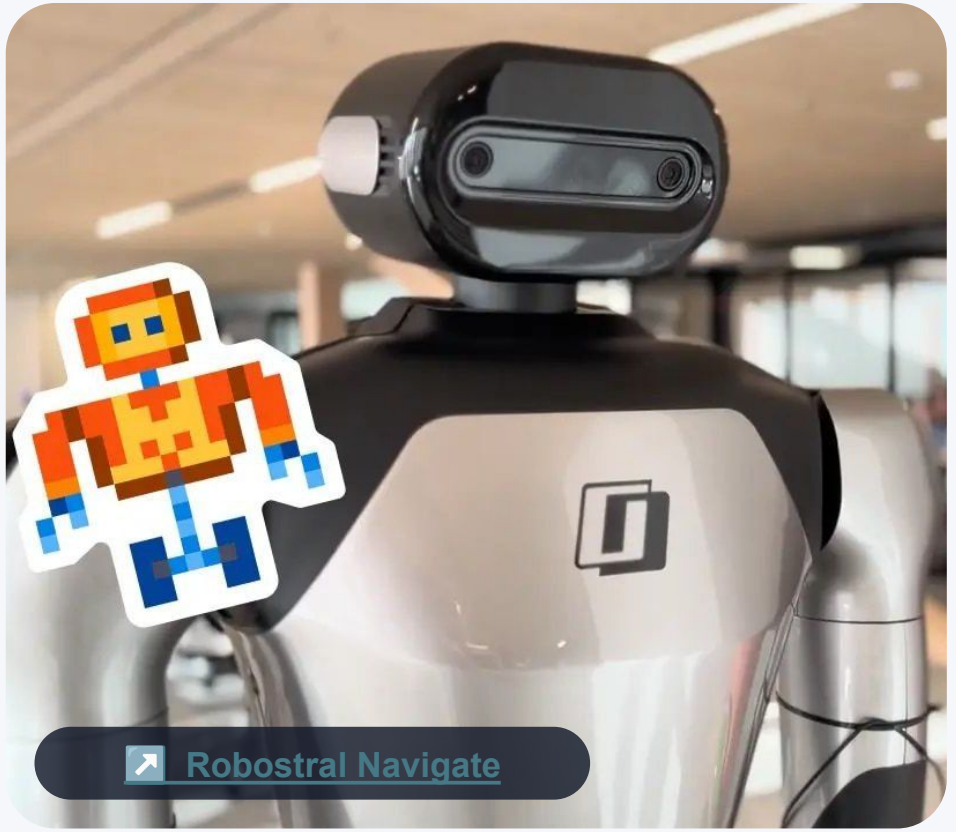
Sources: Mistral AI, Mistral Large 4 announcement (Oct. 2026) · OfficeChai / Artificial Analysis Intelligence Index v4.3.2

Why robotics cares: same team ships Robostral → a European robot brain, no US or Chinese API

< Mistral goes physical />

One camera.
No LiDAR.
A whole warehouse.

Robostral Navigate, July 2026
Mistral’s first robotics model · + Emmi AI acquisition



Sources: Mistral AI (July 2026)

< Mistral × defense: one European AI stack />

Models

Le Chonk, open weights trained in Europe



Robots

Robostral Navigate camera-only navigation

Army

FR

Framework deal, Jan. 2026
all forces + CEA, ONERA
French servers only
supervised by AMIAD



Defense

Helsing partnership (Feb. 2025): AI for defense systems

US labs won’t lend their best AI to our armies. Europe needs its own, and today that is mostly Mistral.

Sources: Blog Nouvelles Technologies / Acteurs Publics (Mistral framework agreement, Jan. 2026) · MLex (Helsing × Mistral, Feb. 2025) · Mistral AI

< Defense tech: Europe’s new robotics money />



Helsing

Germany

$1.8B

raised · $18B valuation

HX-2 strike drone, AI battle software



Harmattan AI

France

$200M

Dassault-led · 1st FR defense unicorn

AI drones, 1,000 ordered by the French army



Quantum Systems

Germany

$1.2B

raised · $8B valuation

surveillance drones



Stark

Germany

€500M

raised · €3.2B valuation

unmanned systems

Sources: Defense News (July 2026) · Sifted / Bloomberg (Harmattan AI)

Parrot: Pivoting from Consumer to Defense Drones

< French forces: AI proposes, a human fires />



~20
ground & air robots cooperating

2027
first robotised combat unit

1
human takes the fire decision

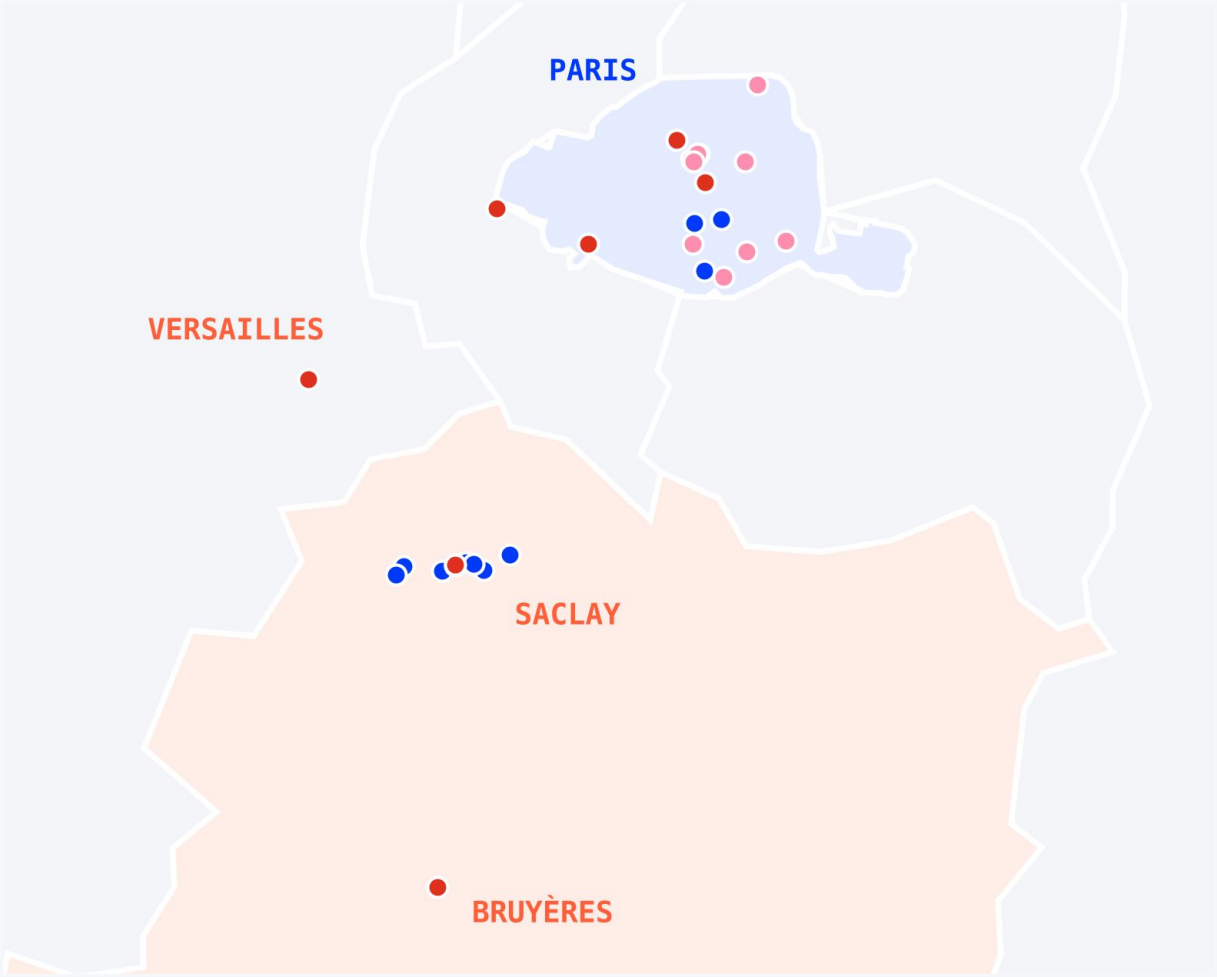


Flew 2 sovereign AIs (Dassault, Thales cortAIx, Sept. 2026)
Rafale + uncrewed wingman: first flight 2028

AMIAD: the ministry's AI agency, co-leads Pendragon
Industry: Safran · KNDS · Thales · Harmattan · Helsing

Sources: defense.gouv.fr (Pendragon, photo © Ministère des Armées) · Option Finance · Migflug · Rafale photo: Laurent Errera, CC BY-SA 2.0

< Robotics & AI around Paris />



INSTITUTES

ISIR · Inria Paris · Mines Paris
Saclay: CEA-List · ENSTA · Télécom Paris
Inria Saclay · LISN · Polytechnique
CentraleSupélec · ONERA

BIG COMPANIES

Mistral AI (HQ + Bruyères data centre)
Meta FAIR · Google DeepMind
Safran · Thales · Dassault Aviation · KNDS

STARTUPS

Wandercraft · Enchanted Tools (Mirokaï)
Bleu Robotics · Robeauté · Ganymed
Fuzzy Logic · Stanley Robotics · Inbolt
Station F (hub)

NOT ON THE MAP

Paris, no public address: Hugging Face · Kyutai · H Company
Genesis AI · UMA · Moon Surgical · Squaremind · Harmattan AI · Helsing
Elsewhere: Pollen (Bordeaux) · Exotec (Lille) · NAVER Labs (Grenoble)

{EPITECH}

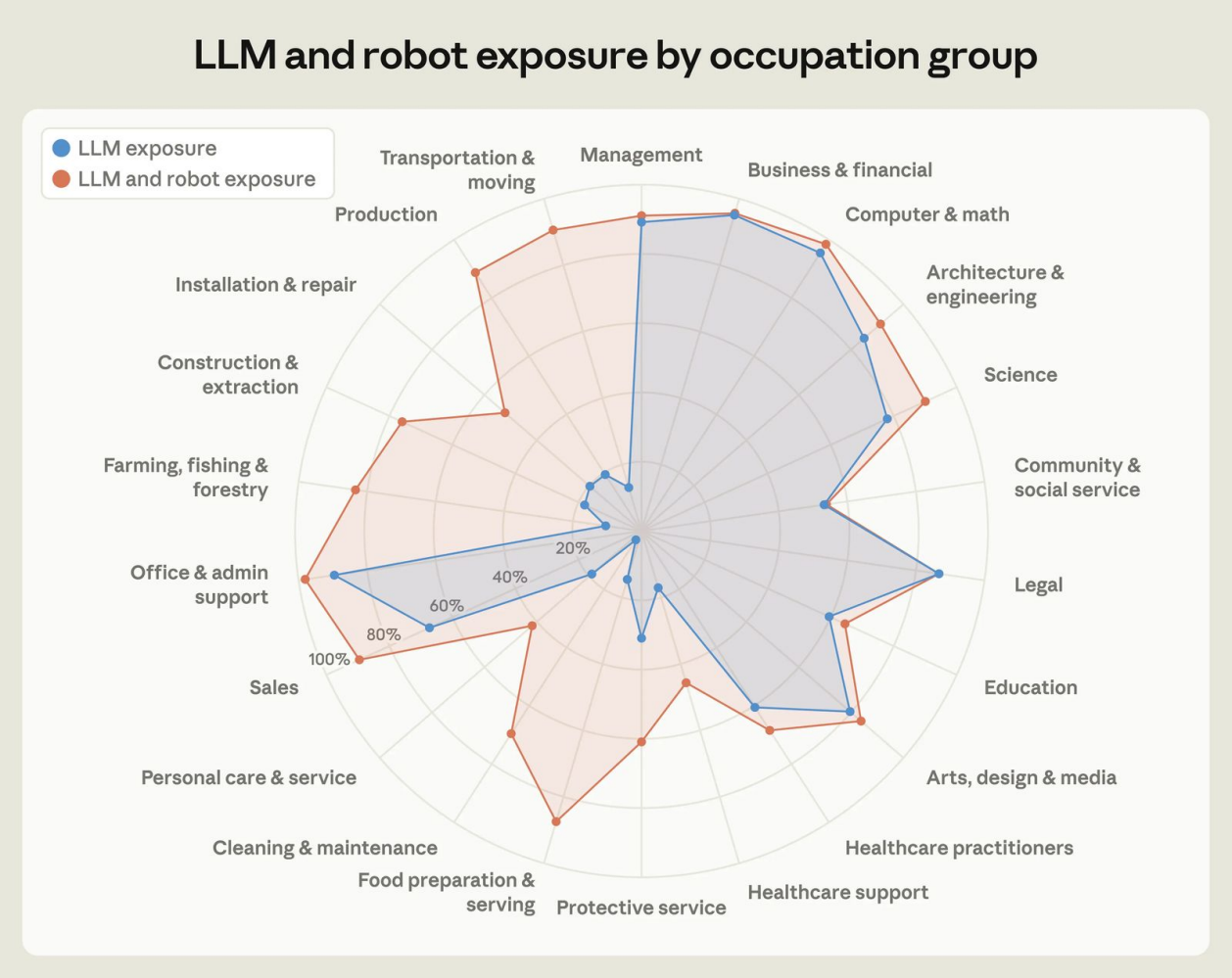
<SINCE 1999/>

< Part 1 · 5/5 />

Ethics: jobs, energy, war

{ <TECH TOGETHER TOMORROW> }

< Will AI or robots take your job? />



ROBOTS CAN DO

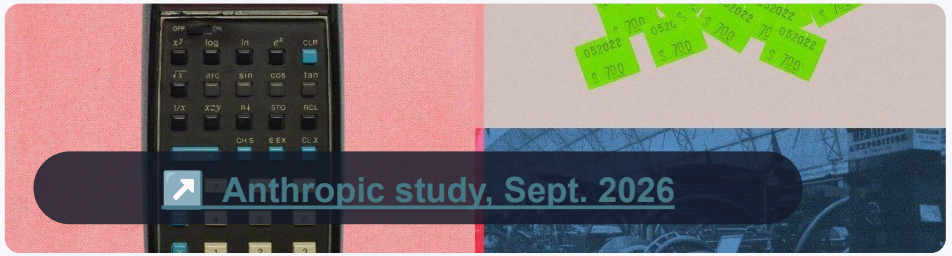
74%

of physical tasks (in limited settings)

BUT ARE CHEAPER THAN HUMANS FOR

0.3%

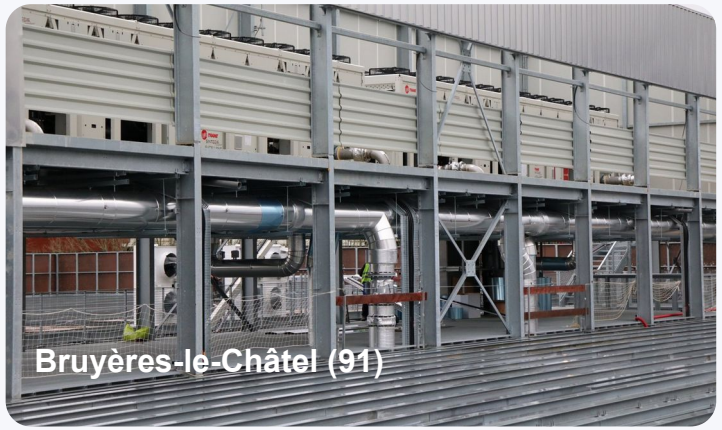
of tasks · ~40 years to reach 10%



TEAMS POLL › Your future job: safe / LLMs / robots / both?

< Who pays the energy bill? />


Cattenom nuclear plant


Bruyères-le-Châtel (91)

×2 **data-centre electricity by 2030 (IEA)**

Campus IA, Fouju (77)

1.4 GW grid link · up to 10.5 TWh/year
more than the whole rail network (SNCF Réseau, 7 TWh)
UAE fund MGX + Bpifrance · NGOs in court (Aug. 2026)

Mistral, Bruyères-le-Châtel (91)

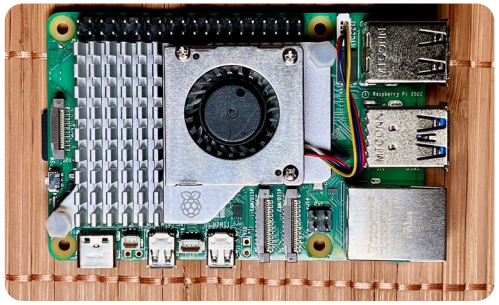
Up to 44 MW · 13,800 NVIDIA GPUs
$830M of debt (Mar. 2026) · run by Eclairion

Sources: IEA · Compare the Cloud (Campus IA) · Reuters (Mistral) · Photos: Wikimedia Commons, gje (CC BY 4.0) · © Région Île-de-France

“AI factories” in Europe: good news or bad news?

< Robot brains got more expensive />

AI data centres eat the memory: your robot pays



Raspberry Pi 5 · 4GB

70 € → 137 €

+95%

2023 → Nov. 2026



Jetson Orin Nano Super

≈ 285 € → 519 €

+82%

Dec. 2024 → Nov. 2026



Jetson AGX Orin 64GB

≈ 2,400 € → 4,013 €

+67%

2022 → Nov. 2026



Jetson AGX Thor

≈ 3,580 € → 6,295 €

+76%

Aug. 2025 → Nov. 2026

Why: RAM price (DDR5 2×32 GB kit)



×6

Sources: Kubii (Pi 5) · welectron (Orin Nano Super) · RS Online (AGX Orin, Thor) · launch ≈ US MSRP + VAT · TechRadar / PCPartPicker (RAM) · photos: NVIDIA, Wikimedia (CC BY)

< Who pulls the trigger? />



DETECT	IDENTIFY	DECIDE	FIRE
AI	**AI**	**HUMAN**	**machine**
cameras, radar	friend or foe?	France: always	only after “yes”

Why it is hard

No link, no human: jamming.

Why it is your topic

Same code as your pick & place.

Photo: FPV drones, Ukraine, ArmiyaInform, CC BY 4.0

TEAMS POLL › Would you work for a defense tech company?

{EPITECH}

<SINCE 1999/>

< Part 2 />

Your projects

{ <TECH TOGETHER TOMORROW> }

< Phase 2 · Control it with your hands or voice />



1

Control it with your hands or voice

Late coming

Control the arm with your hands and fingers.

BJTU

Make the arm dance to a rhythm.
+ Then: high-five detection or a robot that dodges you; speech-to-text, LLM, voice; hand control with MediaPipe (“retargeting”).

Romain Chevalier

Voice commands to trigger robot actions (VOSK), maybe an LLM ?



2

Play a game

Chifoumi

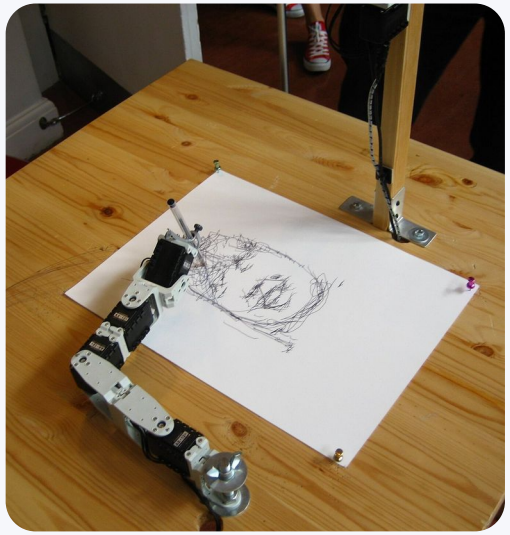
Rock-paper-scissors: OpenCV + MediaPipe hand keypoints, real-time demo OK, GUI with the camera, robot animation OK in simulation.
Next: maybe chess.

Pochart

Tic-tac-toe played with your voice; the robot draws the crosses on the grid with a pencil.

Samuel Valentin

Connect 4, with a way to grab the pieces easily.
Perception: maybe YOLO.



3

Draw & write

Kremlin Bicêtre Dynamics

Redraw a model with a pen.

J’en sais rien

Write letters and sentences, ouija board. Or unplug things.

< Phase 2 · Sort & build />



4

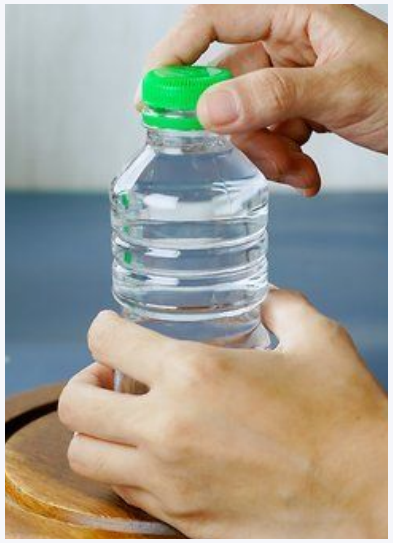
Sort & build

Singenieur

Several balls of different colours: sort them or separate them.

Robotics Chaadi

Stack cubes, build shapes like pixel art, reproduce a drawing with your cubes.



5

Real-world tasks

Neo Grandval

Detect the bottle, grab it and lift it. Then try to remove or put back its cap.

Hal-900

Mount the arm on wheels: navigation with a Kiwi base, or a small car.

< Stats />



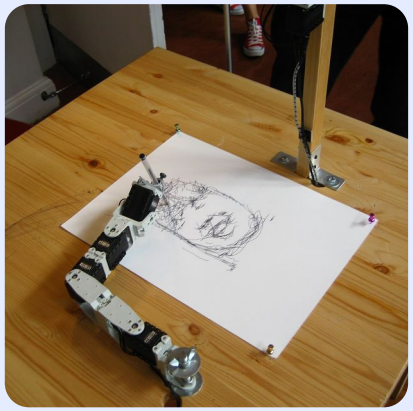
13

groups · 5 themes



3

games: tic-tac-toe,
Connect 4, chifoumi



3

groups draw
or write



3 + 3

control by hand
· by voice

< Follow-ups: order, 10 min each />

MORNING · 11:30

11:30	Late coming
11:40	BJTU
11:50	Hal-900

Lunch break 12:00 – 13:30

AFTERNOON · 13:30

13:30	Singenieur
13:40	J’en sais rien
13:50	Robotics Chaadi
14:00	Romain Chevalier
14:10	Neo Grandval
14:20	Kremlin Bicêtre Dynamics
14:30	Pochart
14:40	Samuel Valentin
14:50	GuiZac
15:00	Chifoumi